

Conjectures for Primitive Pythagorean Triples

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May 1, 2026

1 Overview

We discuss the set $\mathcal{H}$ of primitive Pythagorean hypotenuses – the “hypotenuse” values that arise from primitive Pythagorean triples (right triangles with integer sides sharing no common factor) – recall the classical theorems describing its structure, and state an open conjecture about the cumulative parity of its prime elements.

A related set, $\mathcal{H}_p$, is the subset of $\mathcal{H}$ that are primes. That is, primitive Pythagorean hypotenuses that are prime.

Two of the three observations below turn out to be classical theorems — we recall them for orientation — while the third remains an open conjecture of mine, which is the focus of this note.

1. **Partition of $\mathcal{H}$** (theorem): $\mathcal{H}$ is *partitioned* into two disjoint infinite sets: the set of all powers of Pythagorean primes, and the set of all hypotenuses arising from more than one primitive Pythagorean triple.
2. **Infinitude of $\mathcal{H}_p$** (Dirichlet): The set of Pythagorean primes is infinite.
3. $\mathcal{H}_p \sim \mathbb{P}$ (conjecture): $\mathcal{H}_p$ is *similar* to and as *complex* as the primes when measured by certain cumulative-parity metrics.

2 Definitions

After a few definitions we describe a conjecture regarding Primitive Pythagorean Triples. We use standard notation, associating the natural numbers with the symbol $\mathbb{N}^1$ and we use the symbol $\mathbb{P}$ to denote the set of prime numbers.

Definition 1. *Pythagorean Triples*, denoted $\mathcal{T}$, are triples of numbers representing the lengths of the sides of right triangles where all the sides are integers.² Specifically,

$$\mathcal{T} = \{(x, y, z) \mid (x, y, z) \in \mathbb{N}^3 \wedge x^2 + y^2 = z^2\}. \quad (1)$$

Definition 2. *Primitive Pythagorean Triples*, denoted $\mathcal{T}_p$, are Pythagorean triples whose legs share no common factor:³

$$\mathcal{T}_p = \{(x, y, z) \mid (x, y, z) \in \mathbb{N}^3 \wedge \gcd(x, y) = 1 \wedge x^2 + y^2 = z^2\}. \quad (2)$$

Definition 3. *Primitive Pythagorean Hypotenuses*, denoted $\mathcal{H}$, are defined by:

$$\mathcal{H} = \{z \mid \exists (x, y) \in \mathbb{N}^2, (x, y, z) \in \mathcal{T}_p\}. \quad (3)$$

This is the set of all possible hypotenuses of primitive Pythagorean triples.

Definition 4. *Pythagorean Primes*, denoted $\mathcal{H}_p$, are defined by:

$$\mathcal{H}_p = \{z \mid \exists (x, y) \in \mathbb{N}^2, (x, y, z) \in \mathcal{T}_p \wedge z \in \mathbb{P}\}. \quad (4)$$

This is the set of primes that occur as hypotenuses of primitive Pythagorean triples.

Definition 5. *Duplicate Primitive Hypotenuses*, denoted $\mathcal{H}_d$, are defined by:

$$\mathcal{H}_d = \{z \mid \exists (x_1, x_2, y_1, y_2) \in \mathbb{N}^4, (x_1, y_1, z) \in \mathcal{T}_p \wedge (x_2, y_2, z) \in \mathcal{T}_p \wedge \{x_1, y_1\} \neq \{x_2, y_2\}\}. \quad (5)$$

This is the set of primitive hypotenuses that arise from more than one *distinct* primitive Pythagorean triple (i.e. not merely a swap of legs).

Definition 6. *Primitive Hypotenuses Powers*, denoted $\mathcal{H}_u$, are defined by:

$$\mathcal{H}_u = \{z \mid \exists (p, n) \in \mathcal{H}_p \times \mathbb{N}, z = p^n\}. \quad (6)$$

This is the set of all powers of Pythagorean primes.

Definition 7. The *Prime Parity* function, Π , is defined on any prime p with $p \geq 5$ by:

$$\Pi(p) = \begin{cases} 1 & \exists n \in \mathbb{N}, p = 6n + 1 \\ -1 & \exists n \in \mathbb{N}, p = 6n - 1 \end{cases} \quad (7)$$

Definition 8. The *First Deviation* function, Δ_p , is defined by:⁴

$$\Delta_p(n) = \min_m \left\{ m \mid n \leq \left(\sum_{i=1}^m \Pi(\mathcal{H}_p[i]) - \sum_{i=3}^{m+2} \Pi(\mathbb{P}[i]) \right) \right\} \quad (8)$$

¹We assume that $\mathbb{N}$ does *not* include 0.

²Triples are taken as ordered, so $(3, 4, 5)$ and $(4, 3, 5)$ are distinct elements of $\mathcal{T}$.

³Since any common divisor of x and y also divides z , $\gcd(x, y) = 1$ is equivalent to $\gcd(x, y, z) = 1$, the usual primality condition.

⁴If for a given n , $\left\{ m \mid n \leq \left(\sum_{i=1}^m \Pi(\mathcal{H}_p[i]) - \sum_{i=3}^{m+2} \Pi(\mathbb{P}[i]) \right) \right\} = \emptyset$, we set $\Delta_p(n) = \infty$.

Notes:

- The subscript u in the name $\mathcal{H}_u$ is not yet justified.
- $\mathcal{H}_p$ is a *proper* subset of the primes: $\mathcal{H}_p \subset \mathbb{P}$.
- If the sets above are treated as lists, they are indexed in sorted order.
- Clearly, Δ_p is a non-decreasing function.

First we list a few facts and state a couple of known theorems before proceeding:

1. The Pythagorean primes are of the form $4n+1$. This is a special case of a more general theorem proven by Pierre de Fermat, known as Fermat's theorem on sums of two squares. According to this theorem, an odd prime number p can be expressed as the sum of two squares if and only if p is congruent to 1 modulo 4, i.e., $p = 4n + 1$ for some positive integer n .
2. Moreover, the distribution of primes of the form $4n + 1$ has been studied extensively. The German mathematician Peter Gustav Lejeune Dirichlet proved that there are infinitely many primes of the form $4n + 1$, a result known as Dirichlet's theorem on arithmetic progressions. This theorem, which generalizes the case of Pythagorean primes, states that for any two positive coprime numbers a and d , there are infinitely many primes of the form $a + nd$, where n is a non-negative integer.
3. In the case of Pythagorean primes, we can take $a = 1$ and $d = 4$ to get the form $4n + 1$. Dirichlet's theorem thus guarantees that there are infinitely many Pythagorean primes. This is a profound result, as it extends the prime number theorem, which states that there are infinitely many prime numbers, to arithmetic progressions.

From these results we get the following characterization of Pythagorean primes:

Theorem 1. $\mathcal{H}_p = \{p \mid p \in \mathbb{P} \wedge p \equiv 1 \pmod{4}\}$.

And the following structure theorem for Pythagorean hypotenuses:

Theorem 2. *The set $\mathcal{H}$ has a unique factorization (up to ordering) in $\mathcal{H}_p$: each element of $\mathcal{H}$ can be uniquely written as a product of powers from $\mathcal{H}_p$,*

$$\forall h \in \mathcal{H}, \exists! N \in \mathbb{N}, \exists! (\mathbf{p}, \mathbf{k}) \in (\mathcal{H}_p^N \times \mathbb{N}^N) : h = \prod_{i=1}^N p_i^{k_i}, \quad (9)$$

where uniqueness is up to permutation of the index i .

A classical consequence is the following partition theorem, which I had originally stated as a conjecture but which follows from the count of primitive representations of n as a sum of two squares:

Theorem 3. *The set $\mathcal{H}$ is partitioned by $\mathcal{H}_d$ and $\mathcal{H}_u$:*

1. $\mathcal{H} = \mathcal{H}_d \cup \mathcal{H}_u$;
2. $\emptyset = \mathcal{H}_d \cap \mathcal{H}_u$;
3. $|\mathcal{H}_d| = \infty$.

Proof sketch. For $n \in \mathcal{H}$, the number of primitive Pythagorean triples with hypotenuse n equals 2^{k-1} , where k is the number of distinct prime factors of n (all of which lie in $\mathcal{H}_p$ by Theorem 2). This count is a classical consequence of the unique factorization of n in the Gaussian integers. Hence n has exactly one primitive representation iff $k = 1$, i.e. iff n is a power of a single Pythagorean prime ($n \in \mathcal{H}_u$); otherwise $n \in \mathcal{H}_d$. This gives the partition. The infinitude of $\mathcal{H}_d$ follows from the infinitely many distinct products $p \cdot q$ with $p, q \in \mathcal{H}_p$ and $p \neq q$ (Dirichlet). $\square$

3 McIntire’s Pythagorean Triple Conjectures

The remaining open conjecture concerns the *cumulative parity* of $\mathcal{H}_p$ — the running sum $\sum_{i=1}^n \Pi(\mathcal{H}_p[i])$ — and how it compares to the corresponding sum over all primes ≥ 5 .

Conjecture 1. *The cumulative parity of $\mathcal{H}_p$ exhibits the following non-trivial behavior:*

1. $\lim_{n \rightarrow \infty} \frac{\sum_{i=1}^n \Pi(\mathcal{H}_p[i])}{n} = 0;$
2. $\forall n \in \mathbb{N}, \Delta_p(n) < \infty;$
3. $\lim_{n \rightarrow \infty} \frac{\Delta_p(n)}{n^{\ln(2\pi)}} = 41.^5$

The three sub-conjectures together suggest that the Pythagorean primes are “as complex as” the full set of primes when measured by their parity behavior. Sub-conjecture 1 says the ± 1 values $\Pi(\mathcal{H}_p[i])$ average to zero, so $\mathcal{H}_p$ is spread evenly between residues 1 and $-1 \pmod{6}$ in the long run, mirroring $\mathbb{P}$. Numerically, however, the cumulative parity of $\mathcal{H}_p$ tends to run lower than that of $\mathbb{P}$; sub-conjectures 2 and 3 quantify how often and how far the $\mathcal{H}_p$ sum nonetheless overtakes the $\mathbb{P}$ sum, asserting that deviations of any size n do occur ($\Delta_p(n) < \infty$) and that they grow at the polynomial rate $n^{\ln(2\pi)}$. This “mixing” is what we mean by saying $\mathcal{H}_p$ is non-trivial in a way similar to $\mathbb{P}$.

⁵The exponent $\ln(2\pi) \approx 1.838$ and the constant 41 are based on numerical fitting; we do not have a heuristic derivation. Readers should treat the precise values as empirical, subject to revision as more data becomes available.

A A formally verified Lean 4 proof of Theorem 3

The proof below has been mechanically checked by Lean 4 (toolchain `leanprover/lean4:v4.29.1`) against Mathlib. The full project lives at `lean/PythHyp/` in this repository; the proof file is `lean/PythHyp/PythHyp/Basic.lean`.

The decomposition is honest: three classical results — explicitly named (F), (G-easy), and (G-uniq) — are stated as `axioms` rather than proved in this file, because their formalization, while certainly possible in Mathlib, is a substantial project of its own. They are classical theorems, not conjectures:

- (F) *Fermat's characterization.* A positive integer $z > 1$ is a primitive Pythagorean hypotenuse iff every prime factor of z is $\equiv 1 \pmod{4}$. The prime case is in Mathlib as `Nat.Prime.sq_add_sq`; the general case follows from multiplicativity of the norm in $\mathbb{Z}[i]$.
- (G-easy) *Multiple factors yield duplicates.* If $z \in \mathcal{H}$ has at least two distinct prime factors, then z admits at least two distinct unordered primitive representations $a^2 + b^2 = z^2$. This is the easy half of the classical 2^{k-1} count and follows from a single non-trivial factorization of z in $\mathbb{Z}[i]$.
- (G-uniq) *Prime powers are unique.* If $z = p^n$ for a Pythagorean prime p and $n \geq 1$, then z admits only one unordered primitive representation. This is the harder half of the Gauss count.

Dirichlet's theorem on primes $\equiv 1 \pmod{4}$ is *not* an axiom in this proof: it is invoked directly as the Mathlib lemma `Nat.infinite_setOf_prime_modEq_one`.

Modulo the three classical inputs (F), (G-easy), (G-uniq), every step of Theorem 3 — parts (a), (b), and (c) — is fully formalized, with no `sorry`. Lean reports a clean build.

```
import Mathlib.NumberTheory.PythagoreanTriples
import Mathlib.NumberTheory.SumTwoSquares
import Mathlib.NumberTheory.Zsqrtd.GaussianInt
import Mathlib.NumberTheory.PrimesCongruentOne
import Mathlib.Data.Nat.PrimeFin

open Set

namespace PrimPythHyp

/-- z is a primitive Pythagorean hypotenuse. -/
def IsHyp (z : ℕ) : Prop :=
  ∃ a b : ℕ, 0 < a ∧ 0 < b ∧ Nat.gcd a b = 1 ∧ a^2 + b^2 = z^2

def H : Set ℕ := { z | IsHyp z }
def Hp : Set ℕ := { z | z ∈ H ∧ z.Prime }

/-- Hypotenuses with at least two distinct unordered primitive
    representations. -/
def Hd : Set ℕ :=
  { z | ∃ a₁ b₁ a₂ b₂ : ℕ,
    0 < a₁ ∧ 0 < b₁ ∧ Nat.gcd a₁ b₁ = 1 ∧ a₁^2 + b₁^2 = z^2 ∧
    0 < a₂ ∧ 0 < b₂ ∧ Nat.gcd a₂ b₂ = 1 ∧ a₂^2 + b₂^2 = z^2 ∧
```

```

({a1, b1} : Finset ℕ) ≠ {a2, b2} }

/-- Powers of Pythagorean primes. -/
def Hu : Set ℕ := { z | ∃ p ∈ Hp, ∃ n : ℕ, 0 < n ∧ z = p^n }

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-- Classical building blocks (axiomatized; these are real theorems)
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/-- (F) Fermat's two-squares + multiplicativity in ℤ[i]. -/
axiom mem_H_iff_factors {z : ℕ} (hz : 1 < z) :
  z ∈ H ↔ ∀ p ∈ z.primeFactors, p % 4 = 1

/-- (G-easy) ≥ 2 distinct prime factors ⇒ ≥ 2 distinct primitive reps. -/
axiom multiple_factors_yields_Hd {z : ℕ} (hz : z ∈ H)
  (hk : 2 ≤ z.primeFactors.card) : z ∈ Hd

/-- (G-uniq) prime power ⇒ only one primitive rep. -/
axiom prime_power_not_in_Hd {p : ℕ} (hp : p ∈ Hp) {n : ℕ} (hn : 0 < n) :
  p^n ∉ Hd

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-- Helper
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lemma one_lt_of_in_H {z : ℕ} (hz : z ∈ H) : 1 < z := by
  obtain ⟨a, b, ha, hb, _, hab⟩ := hz
  have h1a : 1 ≤ a^2 := Nat.one_le_iff_ne_zero.mpr (pow_ne_zero 2 ha.ne')
  have h1b : 1 ≤ b^2 := Nat.one_le_iff_ne_zero.mpr (pow_ne_zero 2 hb.ne')
  have h2 : 2 ≤ z^2 := by rw [← hab]; omega
  rcases Nat.lt_or_ge z 2 with hlt | hge
  · interval_cases z <;> simp_all
  · omega

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-- Theorem 3
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/-- (a) H = Hd ∪ Hu. -/
theorem H_eq_Hd_union_Hu : H = Hd ∪ Hu := by
  ext z
  refine ⟨fun hz => ?, ?⟩
  · have hz1 : 1 < z := one_lt_of_in_H hz
    have hz_ne : z ≠ 0 := by omega
    have hcard_pos : 1 ≤ z.primeFactors.card := by
      have : z.primeFactors.Nonempty := by
        rw [Nat.nonempty_primeFactors]; exact hz1
      exact Finset.card_pos.mpr this
    by_cases hk : z.primeFactors.card = 1
    · -- One prime factor: z = p^n.
      right
      obtain ⟨p, hp_eq⟩ := Finset.card_eq_one.mpr hk
      have hp_in : p ∈ z.primeFactors := by
        rw [hp_eq]; exact Finset.mem_singleton_self p
      have hp_prime : p.Prime := Nat.prime_of_mem_primeFactors hp_in
      have hp_dvd : p | z := Nat.dvd_of_mem_primeFactors hp_in
      set n := z.factorization p
      have hn_pos : 0 < n :=
        hp_prime.factorization_pos_of_dvd hz_ne hp_dvd
      have hz_eq : z = p ^ n := by

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    have hprod : z.factorization.prod (· ^ ·) = z :=
      Nat.prod_factorization_pow_eq_self hz_ne
    have hsupp : z.factorization.support = {p} := by
      rw [Nat.support_factorization, hp_eq]
      rw [← hprod, Finsupp.prod, hsupp, Finset.prod_singleton]
    refine ⟨p, ⟨?, hp_prime⟩, n, hn_pos, hz_eq⟩
    rw [mem_H_iff_factors hp_prime.one_lt]
    intro q hq
    rw [hp_prime.primeFactors, Finset.mem_singleton] at hq
    rw [hq]
    exact (mem_H_iff_factors hz1).mp hz p hp_in
  · -- ≥ 2 prime factors.
    left
    have hk2 : 2 ≤ z.primeFactors.card := by omega
    exact multiple_factors_yields_Hd hz hk2
· rintro (hd | hu)
· obtain ⟨a, b, _, _, ha, hb, hgcd, heq, _⟩ := hd
  exact ⟨a, b, ha, hb, hgcd, heq⟩
· obtain ⟨p, hpH, n, hn, rfl⟩ := hu
  have hp_prime : p.Prime := hpH.2
  have hpn_gt1 : 1 < p^n := one_lt_pow0 hp_prime.one_lt hn.ne'
  rw [mem_H_iff_factors hpn_gt1]
  intro q hq
  have hpf : (p^n).primeFactors = {p} :=
    Nat.primeFactors_prime_pow hn.ne' hp_prime
  rw [hpf, Finset.mem_singleton] at hq
  rw [hq]
  exact (mem_H_iff_factors hp_prime.one_lt).mp hpH.1 p
    hp_prime.mem_primeFactors_self

/-- (b) Hd ∩ Hu = ∅. -/
theorem Hd_inter_Hu : Hd ∩ Hu = ∅ := by
  ext z
  simp only [mem_inter_iff, mem_empty_iff_false, iff_false, not_and]
  rintro hd ⟨p, hpH, n, hn, rfl⟩
  exact prime_power_not_in_Hd hpH hn hd

/-- (c) Hd is infinite. -/
theorem Hd_infinite : Hd.Infinite := by
  set S : Set ℕ := {p | p.Prime ∧ p ≡ 1 [MOD 4]} \ {5}
  have h_pyth : Set.Infinite {p : ℕ | p.Prime ∧ p ≡ 1 [MOD 4]} :=
    Nat.infinite_setOf_prime_modEq_one (by norm_num)
  have hS_inf : S.Infinite := h_pyth.diff (Set.finite_singleton _)
  apply Set.infinite_of_injOn_mapsTo
    (s := S) (t := Hd) (f := fun p => 5 * p)
  · intro p1 _ p2 _ h; dsimp only at h; omega
  · intro p hpS
    obtain ⟨(hp_prime, hp_mod), hp_ne⟩ := hpS
    have hp_ne_5 : p ≠ 5 := fun h => hp_ne (by simp [h])
    have h5_prime : Nat.Prime 5 := by decide
    have hp_pos : 0 < p := hp_prime.pos
    have hp_ne0 : p ≠ 0 := hp_pos.ne'
    have h5p_gt1 : 1 < 5 * p := by
      have : 1 < p := hp_prime.one_lt; omega
    have h5p_in_H : 5 * p ∈ H := by
      rw [mem_H_iff_factors h5p_gt1,
        Nat.primeFactors_mul (by norm_num) hp_ne0,
        h5_prime.primeFactors, hp_prime.primeFactors]
    intro q hq
    rw [Finset.mem_union, Finset.mem_singleton,

```

```

    Finset.mem_singleton] at hq
  rcases hq with rfl | rfl
  · decide
  · exact hp_mod
have hcard : 2 ≤ (5*p).primeFactors.card := by
  have h5_in : (5 : ℕ) ∈ (5*p).primeFactors :=
    Nat.mem_primeFactors.mpr ⟨h5_prime, ⟨p, rfl⟩, by omega⟩
  have hp_in : p ∈ (5*p).primeFactors :=
    Nat.mem_primeFactors.mpr ⟨hp_prime, ⟨5, by ring⟩, by omega⟩
  have hsub : ({5, p} : Finset ℕ) ⊆ (5*p).primeFactors := by
    intro x hx
    rw [Finset.mem_insert, Finset.mem_singleton] at hx
    rcases hx with rfl | rfl <|> assumption
  have h2 : ({5, p} : Finset ℕ).card = 2 := by
    rw [Finset.card_insert_of_notMem
      (by simp [Ne.symm hp_ne_5]), Finset.card_singleton]
  calc 2 = _ := h2.symm
  _ ≤ _ := Finset.card_le_card hsub
exact multiple_factors_yields_Hd h5p_in_H hcard
· exact hS_inf

/-- Theorem 3: the partition theorem. -/
theorem H_partition :
  H = Hd ∪ Hu ∧ Hd ∩ Hu = ∅ ∧ Hd.Infinite :=
  ⟨H_eq_Hd_union_Hu, Hd_inter_Hu, Hd_infinite⟩

end PrimPythHyp

```

Once (F), (G-easy), (G-uniq) are themselves formalized, replacing the `axiom` declarations with `theorem` proofs, this file would become a self-contained, machine-checked proof of Theorem 3 with no classical inputs assumed beyond what is already in Mathlib.